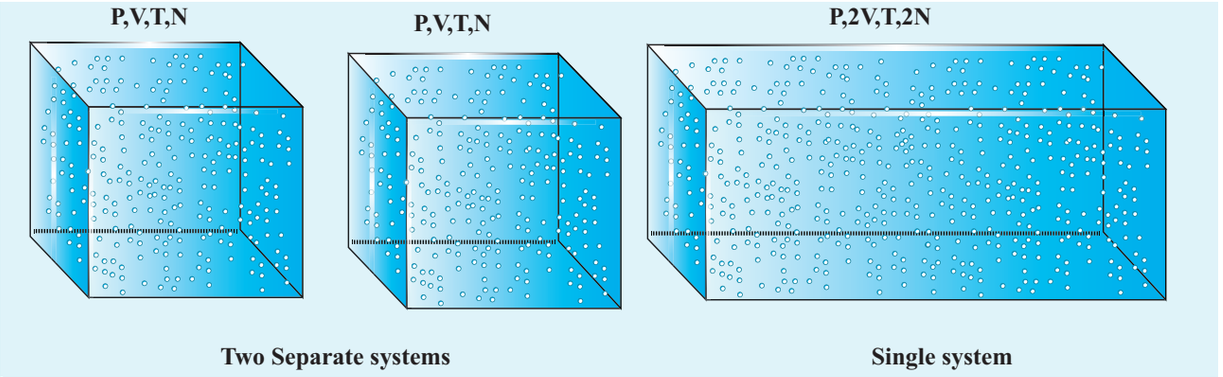


Figure 8.2 Ideal gas law

For this combined system, V becomes 2V, so C should also double to match with the ideal gas equation $\frac{P(2V)}{T} = 2C$. It implies that C must depend on the number of particles in the gas and also should have the dimension of $\left[\frac{PV}{T}\right] = \text{JK}^{-1}$. So we can write the constant C as k times the number of particles N.

Here k is the Boltzmann constant (1.381×10^{-23} JK^{-1}) and it is found to be a universal constant.

So the ideal gas law can be stated as follows

$$PV = NkT \qquad (8.1)$$

The equation (8.1) can also be expressed in terms of mole.

Mole is the practical unit to express the amount of gas. One mole of any substance is the amount of that substance which contains Avogadro number (N_A) of particles (such as atoms or molecules). The Avogadro's number N_A is defined as the number of carbon atoms contained in exactly 12 g of ^{12}C.

Suppose if a gas contains μ mole of particles then the total number of particles can be written as

$$N = \mu N_A \qquad (8.2)$$

where N_A is Avogadro number ($6.023\times10^{23}\text{mol}^{-1}$)

Substituting for N from equation (8.2), the equation (8.1) becomes

$PV = \mu N_A kT$. Here $N_A k = R$ called universal gas constant and its value is 8.314 J /mol. K.

So the ideal gas law can be written for μ mole of gas as

$$PV = \mu RT \qquad (8.3)$$

This is called the equation of state for an ideal gas. It relates the pressure, volume and temperature of thermodynamic system at equilibrium.

EXAMPLE 8.2

A student comes to school by a bicycle whose tire is filled with air at a pressure 240 kPa at 27°C. She travels 8 km to reach the school and the temperature of the bicycle tire increases to 39°C. What is the change in pressure in the tire when the student reaches school?

UNIT-8(XI-Physics_Vol-2).indd 98 02-01-2023 15:25:16